Projectile Motion: Predicting Where the Ball Lands

SYNOPSIS

In M2 N4 you argued that a horizontally launched ball and a dropped ball hit the ground at the same instant, because the horizontal and vertical motions are independent. This lab puts money on that claim. You will launch a ball horizontally from one height and measure where it lands — and that single measurement, combined with the independence claim, is enough to tell you the launch speed. Then you move the launcher to a height you have never used, calculate where the ball will land, **put a cup there**, and fire. Either physics predicts the future or it does not, and you are about to find out in front of witnesses.

Due Sunday, October 11, 11:59 PM | **Report** Full lab report — use the Lab Template | **Pairs with** M2 N4, M2 N6

PERFORMANCE EXPECTATIONS By the end of this lab, you will be able to…

- **PE1** **Treat a two-dimensional motion as two one-dimensional motions** sharing a single clock — constant velocity horizontally, constant acceleration vertically.
- **PE2** **Solve a system of two kinematic equations** by eliminating the shared time variable, and derive $d = v_0\sqrt{(2h/g)}$ yourself rather than quoting it.
- **PE3** **Determine an unmeasured quantity indirectly** — obtain the launch speed from a height and a horizontal distance, neither of which is a speed.
- **PE4** **Propagate uncertainty through a calculation**, so that your launch speed carries an uncertainty derived from the uncertainties in h and d.
- **PE5** **Make a quantitative prediction before taking the data**, then test it — and report the result whether or not it worked.
- **PE6** **Plot theory as a curve and experiment as points on one graph**, and read the pattern in the residuals rather than only the average agreement.

The physics you are using

Sign convention here. The physics is **$a = -g$**: upward is positive, and g is a magnitude that never carries a sign. This lab uses **$g = 9.81\ m/s^2$** throughout — the accepted value at our latitude, the same one Lab 02 compares against. The equations below write g without a minus sign because they compute the *magnitude* of the drop below the launch point as time grows; the ball is still accelerating downward at $a = -g$. Quote 9.81 in your report and keep it consistent everywhere, including in your spreadsheet.

You built this in Note 4. *Freefall vs. Horizontal Launch* is where the claim was settled: a ball launched horizontally falls in exactly the same time as a ball simply dropped from the same height, because the two motions are independent and the launched ball starts with no vertical velocity at all. That note derives $t = \sqrt{(2h/g)}$ and leaves one question deliberately open — *keep the launch speed and double the height; does the ball land twice as far out?* **This lab is where you answer it, with a steel ball and a ruler.** Bring the prediction you wrote down.

When an object moves near the surface of the Earth in both directions at once, gravity pulls vertically and does nothing at all horizontally. So the horizontal velocity of a projectile never changes, while its vertical velocity changes steadily. That gives you two kinematic equations for the same flight, one per axis, with **t** shared between them:

$$\textbf{horizontal:}\quad d = v_0\, t \quad (a_x = 0)$$
$$\textbf{vertical:}\quad h = \tfrac{1}{2}\, g\, t^2 \quad (v_{0y} = 0)$$

the ball leaves the launcher horizontally, so its initial vertical speed is zero

Two equations, three unknowns — but **t** is shared, and that is the whole trick. Solve the vertical equation for the fall time, substitute it into the horizontal one, and the time drops out:

$$t = \sqrt{(2h / g)} \quad \rightarrow \quad d = v_0 \sqrt{(2h / g)}$$

Do this derivation yourself in your Introduction, line by line. Do not simply quote the result.

Read that equation two ways. If you know **h** and measure **d**, you get v_0 — a speed you never measured directly and have no stopwatch fast enough to catch. And once you know v_0, the same equation runs forward and predicts **d** for any height you like. That round trip, measurement to prediction, is this lab.

What this lab does *not* measure. Put $v_0 = \bar{d}_0\sqrt{(g/2h_0)}$ from your calibration run back into $d = v_0\sqrt{(2h/g)}$, and watch g cancel:

$$d = \bar{d}_0 \sqrt{(h / h_0)}$$

Where this holds: the same launcher at the same launch speed for every height, released horizontally, air resistance ignored. It is the two boxed equations above with v_0 eliminated — nothing new, just the same physics with the calibration folded in.

Your prediction depends on the calibration pair (h_0, $\bar{d}_0$) and the new height, and on nothing else. **So this is not a measurement of *g***, and a disagreement between your prediction and your data cannot be blamed on the value of g you used — it would survive any value at all. What you are testing is the **scaling law**: that d grows as $\sqrt{h}$. Say so in your Introduction. It is also the whole content of the prediction below, and it is what makes discussion question 7 worth answering.

Predict first. You double the launch height. Ignoring air resistance, the horizontal distance the ball travels will...

(A) Double as well.

(B) Increase by a factor of √2, about 1.41.

(C) Quadruple.

(D) Stay the same — the launch speed has not changed.

What you need

- Horizontal ball launcher and a small steel ball or marble
- A shelf unit, a set of tables, or anything that gives you launch heights from about 0.20 m to over 1.00 m — the wider the range of heights, the more clearly the relationship shows
- Two-meter stick or tape measure; a level
- Carbon paper and plain paper; tape

- A plastic cup (not glass or ceramic — you will be aiming a steel ball at it)
- Camera for setup photos and for the Extras
- The **Lab Template** and the **Writing Lab Reports** module

> **Two requirements before you take a single measurement.** First, you must be able to measure the height of the *end of the launcher* — that is where the ball's flight begins, and it is not the same as the height of the shelf. Second, the launcher must be genuinely horizontal; check it with a level rather than by eye. Make sure the ball has a clear path all the way to the floor. And always start the ball at the top of the launcher, every single time — the launcher is designed to give a consistent launch speed, but only if it is loaded consistently.

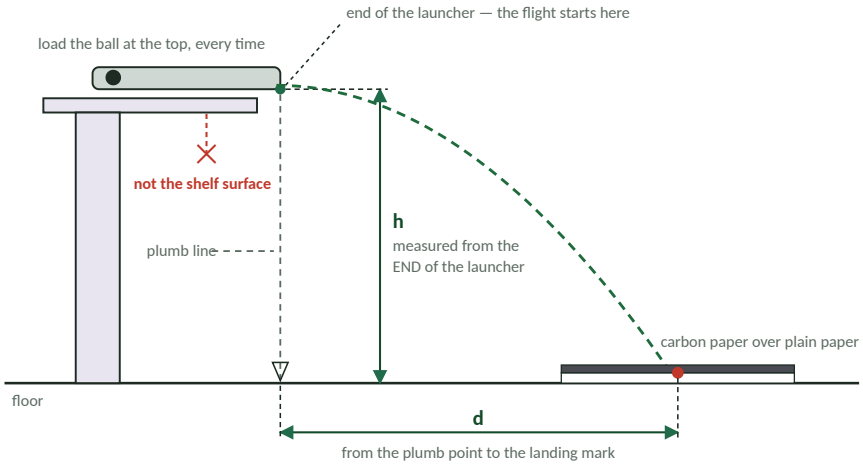


Figure A. Where the two measurements are taken. **h** runs from the *end of the launcher* to the floor — not from the shelf surface, which sits lower and will bias every prediction you make. **d** runs from the point directly below the launcher end, found with a plumb line, to the landing mark. Your own report needs a photo of your apparatus *and* a labelled sketch of your own; this one is not a substitute for either.

Part I — Determining the launch speed

1. Mount the launcher at your first height. Measure **h**, the vertical drop from the end of the launcher to the floor. Record your measurement uncertainty along with the value — half the smallest division on your measuring device is a reasonable starting point.
2. Tape a sheet of plain paper down in the landing area, and lay carbon paper over it, carbon side down. Each landing leaves a small mark on the plain sheet, which you can measure afterward at your leisure. This is much more accurate than trying to watch where the ball hits.
3. Place the ball at the top of the launcher and release it. Repeat for **at least five trials** — five is the number the rubric expects.
4. Measure **d** for each mark: the horizontal distance from the point *directly below the end of the launcher* to the landing mark. Drop a plumb line if you need to find that point honestly.
5. Compute the mean horizontal distance and its standard error of the mean.

Getting v_0 and its uncertainty

Use your average distance and your measured height in the equation you derived:

$$v_0 = \bar{d} / \sqrt{(2h / g)} = \bar{d}\,\sqrt{(g / 2h)}$$

Because v_0 depends on d to the first power and on h to the power −½, the relative uncertainties combine like this:

$$\delta v_0 / v_0 = \sqrt{[\,(\delta d / \bar{d})^2 + (\tfrac{1}{2} \cdot \delta h / h)^2\,]}$$

use the standard error of the mean for δd, and your measurement uncertainty for δh

Notice what that factor of ½ is telling you: an error in the height hurts you only half as much as the same relative error in the distance. That is worth a sentence in your discussion, and it tells you which measurement deserves more of your care.

Show this as a worked sample calculation, with units at every step.

Part II — Predicting the landing position

Now run the equation forward. You know v_0; pick a new height and calculate where the ball will land before you fire it.

1. Move the launcher to a height you have **not** used yet. Measure the new h.
2. **Calculate the predicted landing distance** and write it in your lab notebook *before you launch*. This is the point of the lab — a prediction recorded after the fact is not a prediction.
3. Place the cup so its centre is at your predicted distance. Take one shot and record whether it went in. Report that result honestly whichever way it goes.
4. Whether or not you hit the cup, move the cup aside and use carbon paper to record **at least five landing marks** at this height. Measure them and compute the mean and SEM.
5. Repeat all of the above for **at least five different heights**, including the one you used in Part I. Spread them out across the full range you can reach.

Part III — Theory against data on one graph

Make a single graph with launch height **h** on the horizontal axis and horizontal distance **d** on the vertical axis. It carries two things:

- The **theory**, plotted as a smooth dashed curve: compute $d = v_0\sqrt{(2h/g)}$ at closely spaced heights across your whole range — every 0.05 m works well — using the v_0 you determined in Part I.
- The **experiment**, plotted as discrete markers with vertical error bars: your measured mean distance at each height you actually tested.

> **Do not fit a trendline to the experimental points here.** The curve on this graph is not fitted to your data — it is calculated from a v_0 you measured separately, which is precisely what makes the comparison meaningful. Do not connect your data points dot-to-dot either.

The comparison

For each height, compute the percent difference between your measured and predicted distances:

$$\%\ \text{difference} = |d_{\text{measured}} - d_{\text{predicted}}| \,/\, d_{\text{predicted}} \times 100\%$$

Then look at the *pattern* in those percent differences, not just their average. Are they scattered randomly around zero, or do they trend one way as the height increases? Are they larger at small heights or at large heights? A trend in the residuals means something systematic is going on, and finding it is worth more than a small average error.

Check your predictions

Use this to check the arithmetic in your spreadsheet. Your report still has to show the calculation worked by hand, with units.

TOOL · LAUNCH SPEED AND PREDICTED LANDING DISTANCE

Calibration height h_0 (m)

0.927

Mean measured distance $\bar{d}_0$ (m)

0.475

New launch height h (m)

1.217

Measured distance at that height (m) — optional

leave blank to predict only

Uses g = 9.81 m/s² and assumes the ball leaves the launcher horizontally from rest in the vertical direction.

Additional Questions

These three are required. Answer all three in a section of their own headed **Additional Questions**, in **sentences and paragraphs** — never as a numbered list, and never by restating the question before answering it. This section is worth 5 points on the rubric.

AQ1. Why did we not simply use a stopwatch to measure the time of fall? Estimate the actual fall time for one of your heights and compare it to a realistic human reaction time, so the argument is quantitative rather than hand-waved.

AQ2. Suppose the launcher were not quite horizontal. Treat the two cases separately: tilted slightly *above* horizontal, and tilted slightly *below*. Which way does each error push your measured distance, and which way does it push the v_0 you infer from it?

AQ3. Your predicted curve depends on a single number, v_0, measured at one height. What would change in your analysis if you instead determined v_0 separately at every height? What would you gain, and what would you lose?

Prompts for your Discussion

These are not the Additional Questions. Work them into your **Discussion** section as part of interpreting your results, in sentences and paragraphs. Do not limit yourself to these; add any other insight you have.

I. Was air resistance a factor in this experiment? Do not just assert an answer — argue from what your data would look like if it were. If drag mattered, would your measured distances come in above or below prediction, and would the effect grow or shrink with launch height?

II. What would happen if you repeated the whole experiment with a ping-pong ball instead of a marble? Be specific about which part of the analysis would break first.

III. Does the height of the cup affect how easy it is to hit? Could you drop the ball into a tall narrow pipe standing on the floor using the same predictive equation, or would you need to change something?

IV. Look at how your percent differences vary with launch height. If your measurement of **d** carries a roughly fixed absolute error — a centimetre or two, regardless of the height — what does that do to the *percent* difference at small heights compared to large ones? Does that account for the pattern you actually see, or is something else going on?

> **One thing the rubric will not accept:** "human error." It explains nothing. If your percent difference is under 10%, attribute it to ordinary experimental uncertainty and say so. If it is over 10%, you must identify specific error sources and *quantify* them — a named mechanism with a number attached.

Extras

Post a video to the discussion board showing your most impressive shot. What is the smallest cup you can get the ball into? Say what height you fired from and what you predicted.

✔ Before you submit

Every line here maps to a row on the Lab 03 grading rubric.

FORMAT & FRONT MATTER

- [ ] Lab Template used, with names, title, and date in the heading
- [ ] Abstract: four sentences or fewer — theme, numerical results, percent difference. Nothing else.
- [ ] Introduction contains your own derivation of $d = v_0\sqrt{(2h/g)}$, the variables defined, and a hypothesis you can later confirm or reject

METHODS

- [ ] Equipment described within the procedure, in paragraphs — not as a bare list
- [ ] Procedure in paragraph form, in your own words, detailed enough to follow — not copied from this handout
- [ ] Photo of the apparatus *and* a labelled sketch, with every component labelled and both measurement reference points marked — where h is measured from and to, and where d is measured from and to

RESULTS

- [ ] Table of the five-plus calibration trials at the first height, with mean and SEM
- [ ] Table of all launch heights, with predicted distance, mean measured distance, SEM, and percent difference
- [ ] Every table numbered, captioned, units and consistent significant digits, referenced in the text ("Table 1 shows…")
- [ ] The theory-versus-experiment graph: dashed calculated curve, measured points as markers with error bars, numbered as a Figure with a caption and described in the text
- [ ] A sample calculation, worked step by step with units, for *each* type: fall time, v_0, uncertainty propagation in v_0, predicted distance, mean and SEM, percent difference
- [ ] Whether the ball landed in the cup, stated plainly for each height — including the times it missed

DISCUSSION & CONCLUSION

- [ ] Discussion opens with "we experimentally determined that… ± error"
- [ ] States whether your introduction's hypothesis is confirmed, and argues confidence using the 10% rule
- [ ] The *pattern* in the percent differences across heights is addressed, not just their average
- [ ] All three Additional Questions (AQ1–AQ3) answered in a dedicated section, in paragraphs
- [ ] The phrase "human error" appears nowhere in the report

- [ ] Conclusion: 3–4 sentences — numerical results with percent difference, a definitive verified / not-verified statement, and one concrete way to improve the measurement

REFERENCES

- [ ] One to three references beyond the textbook — the handout, the M2 note pages, the walkthrough videos, and an outside source on projectile motion
- [ ] Author, title, and date given for each — a bare URL is not a citation

0 of 20 checked.

Files for this lab

HANDOUT

This lab, as a PDF

The whole page, printable. The calculator and the checklist only work on the web version.

OPTIONAL

Example workbook (XLSX)

The column layout for the calibration, the per-height predictions and the theory curve, filled with a worked example set so you can see the shape of a finished sheet.

REQUIRED

Lab 03 grading rubric

The rubric page, with Word and PDF copies. What each section is worth. Every line of the submission checklist above maps to a row here.

REQUIRED

Report starter (DOCX)

The Lab Template with this lab's headings and equations already in place. Start here rather than from a blank page.

REFERENCE

Formatting Data Tables for Lab Reports

Table numbering, captions, significant digits, units.

REFERENCE

M2 N4 — Freefall vs. Horizontal Launch

The independence-of-motion argument this lab tests.

Abstract rules, section-by-section guidance and the submission window live in the **Writing Lab Reports** module on Brightspace, alongside the Module 2 Lab folder where you hand this in.

References to get you started

Your references should include material on projectile motion. Your textbook, this handout, and the Module 2 note pages all count, and you need at least one source beyond the textbook. The note this lab measures is Module 2 Note 4 — Freefall vs. Horizontal Launch; the angled-launch case you may want for the Extras is Note 6.

1. [Author's names], *Journal or Book title*, edition, Volume number, page number, (year).

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